Siqi Guo

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Education

Carnegie Mellon University Pittsburgh, PA
Master of Science in Electrical and Computer Engineering; GPA: 3.89/4.0 Expected May 2025

Relevant Courses: Foundations of Computer Systems, Embedded System Software Engineering, Distributed Systems, Autonomous Control System, Storage Systems, Modern Computer Architecture and Design, Intro to Embedded Systems, Multi-Agent Planning and Coordination, Software Development for Computational Science

Tianjin University Tianjin, China
Bachelor of Science in Computer Science and Technology; GPA: 3.84/4.0 June 2023

Relevant Courses: Data Structure, Operating System, Computer Networks, Principles of Database, Parallel Computing, Natural Language Processing, Pattern Recognition and Deep Learning

Research & Intern Experience

Eigenbot - Bio-inspired Distributed Control for Modular Robot Pittsburgh, PA
Biorobotics Lab Research Assistant (Part-time) Dec 2023 - Oct 2024
Advisor: Professor Howie Choset, Research Scientist Lu Li Carnegie Mellon University

- Eigenbot Firmware Development and Testing
  - Developed Eigenbot EEPROM parameters auto-setting tools, which accelerated Hexapod gaits tuning process.
  - Tested the firmware. Redesigned and debugged the module communication protocol to ensure the robot's stability and reliability. Conducted the experiments with Logan, analyzed the gaits and fine-tuned firmware parameters.
  - Integrated Foot Sensor Feedback into the robot's message protocol.
  - Collected EigenBot's data from the real-world using OptiTrack, developing a pipeline to assess metrics like Gait Phase Difference and Leg Stance Duration, benchmarking its performance with centralized predefined gait.

E2Biz SoftTech (Tianjin) Co. Ltd Tianjin, China
Web Developer Intern (Full-time) March 2023 - June 2023

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  - Utilized ASP.NET **MVC** to build the Warehousing Order Processing Unit and the Work Order Management module, designed front-end pages with Razor and **KendoUI**, and implemented controllers to do DB filtering.
  - Designed an image compressor with an overall **85%** compression rate and reduced image upload time by **70%** in the project, Product Line Supermarket System.
  - Cooperated with QA testers in eight function unit tests, providing test cases which increased test coverage by 21%.
  - Launched this new project, ESC5Scaffold, and implemented Department Planning and User & Role Management.
  - Upgraded OurCabin System by enhancing the Bug Tracking module and the FAQ section, and especially adding the Excel exporting function for all forms with **Aspose**.
  - Added the Auto-completion of historical records with **HTTP cookies and sessions** in the searching system which decreased the searching cache memory burden by **approximately 90%**.

Projects

CloudFS, a Deduplicated Cloud File System | CMU 18-746 Course Project Oct 2024 - Dec 2024

- Implemented CloudFS, leveraging the properties of local SSD and cloud storage to make data-placement decisions.
- Identified and eliminated duplicate data with Rabin fingerprinting. Minimized storage usage and optimized data transfer in the form of segments. Designed buffer file to manully handle the file data transfer.
- Implemented the IOCTL functions to support snapshot operations, such as create, restore, install, and uninstall.
- Added in-memory LRU caching to further improve performance and reduce cloud operation costs, reaching **NO.1** on the scoreboard among all students.

Distributed File System Design | CMU 15-640 Course Project Feb 2024 - Apr 2024

- Implemented the serialization protocol for the low-level RPC stub, supporting several RPCs in project's NFS.
- Designed a File-Caching Proxy based on RMI. Implement open-close session semantics for proxy operations, and the LRU replacement policy for cache management. Used version control, check-on-use and last-close-win to make sure the proxy's cache and server freshness.
- Implemented and Tuned a 3-Tier Architecture Scalable Web Service to satisfy dynamic or unexpected workloads by auto-scaling. Then, evaluated the performance of the system by benchmarking and figuring out the bottleneck.
- Designed a Two-phase Commit Protocol to ensure the consistency of the distributed file system. Customized a mechanism to recovering from nodes failures and handling lost messages.

Skills

Programming Languages: C/C++, Python, C#, JavaScript, LaTex, SQL, Verilog, Bash
Frameworks: TensorFlow, PyTorch, ROS, NodeJS, Flask, .NET, React, Flutter
Software and Tools: GIT, Docker, MySQL, SQLServer, Qt Creator, Visual Studio Code, Gem5, PSoC
Platforms: MacOS, Linux, Arduino, GCP